Chapter 3. Sequences of Random Variables

Coin tossing, laws of large numbers, independence, product measures, weak convergence, characteristic functions, the central limit theorem

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3.1 Infinitely many fair coin tosses

Motivation. Limits need all tosses on one space, built from Lebesgue measure via binary expansions.

Ω is the set of outcomes of a **random experiment**; probability attaches a number to every event.

One toss. $\Omega = \{0,1\}$, $\mathcal{A} = \mathcal{P}(\Omega)$, $P(\{0\}) = P(\{1\}) = \frac{1}{2}$.

n **tosses.** $\Omega = \{0,1\}^n$, $\mathcal{A} = \mathcal{P}(\Omega)$ and $P(A) = |A|/2^n$: the **uniform distribution**, each binary word having probability 2^{-n}.

Infinitely many tosses. $\Omega = \{0,1\}^{\mathbb{N}} = \{(x_i)_{i\in\mathbb{N}} \mid x_i \in \{0,1\}\}$ is uncountable; $(x_i) \mapsto \sum_i x_i 2^{-i}$ maps it onto $[0,1]$ (not injectively). Let $\mathcal{A} := \sigma(\mathcal{A}_0)$ with the *cylinder sets*

$$\mathcal{A}_0 := \{B \subset \Omega \mid B = B_0 \times \{0,1\} \times \{0,1\} \times \cdots \text{ for some } n \text{ and } B_0 \subset \{0,1\}^n\},$$

(events depending on finitely many tosses); we need P on $\mathcal{A}$ with $P(x_1 = \bar{x}_1, \dots, x_n = \bar{x}_n) = 2^{-n}$.

Existence via Lebesgue measure. Let $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P}) = ([0,1], \mathcal{B}([0,1]), \lambda)$. On $\Omega = \{0,1\}^{\mathbb{N}}$ let $X_i((x_n)_n) := x_i$ (the i-th toss) and $\mathcal{A} := \sigma(\{X_i = 1\} \mid i \in \mathbb{N})$, which equals $\sigma(\mathcal{A}_0)$. The non-terminating binary expansion of $\tilde{\omega} \in [0,1]$ is

$$\tilde{\omega} = \sum_{i=1}^{\infty} T_i(\tilde{\omega}) 2^{-i}, \qquad T_i := \sum_{k=1}^{2^{i-1}} \mathbb{1}_{\left(\frac{2k-1}{2^i}, \frac{2k}{2^i}\right]},$$

and defines $T : \tilde{\Omega} \to \Omega$, $T(\tilde{\omega}) := (T_1(\tilde{\omega}), T_2(\tilde{\omega}), \dots)$. T misses the sequences ending in zeros (other than $(0,0,\dots)$), i.e. the countably many dyadic rationals.

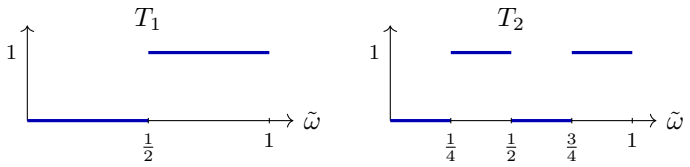


Figure. *Digit maps T_1, T_2; fixing n digits gives an interval of length 2^{-n}.*

Since $T_i = X_i \circ T$, $T^{-1}(\{X_i = 1\}) = \{T_i = 1\}$ is a finite union of intervals, so T is $\tilde{\mathcal{A}}/\mathcal{A}$-measurable (Remark 1.32). $P := \tilde{P} \circ T^{-1}$ is a probability measure on $(\Omega, \mathcal{A})$ (Proposition 2.13), and for $(x_1, \dots, x_n) \in \{0,1\}^n$

$$P(X_1 = x_1, \dots, X_n = x_n) = \tilde{P}\Big(\bigcap_{i\le n} \{T_i = x_i\}\Big) = \tilde{P}(\text{an interval of length } 2^{-n}) = 2^{-n}.$$

So P extends the model for n tosses; uniqueness follows from Theorem 1.9, since the cylinder sets form a $\cap$-stable generator.

Some expectations.

- The i-th toss: $\mathbb{E}[X_i] = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = \frac{1}{2}$.
- Number of successes $S_n := X_1 + \cdots + X_n$: $P(S_n = k) = \binom{n}{k} 2^{-n}$, so $\mathbb{E}[S_n] = \sum_k k\binom{n}{k} 2^{-n} = \frac{n}{2}$; more simply, $\mathbb{E}[S_n] = \sum_i \mathbb{E}[X_i] = \frac{n}{2}$ by linearity.
- Waiting time $S := \min\{n \mid X_n = 1\}$ ($\min \emptyset := \infty$), measurable since $\{S = k\} = \{X_1 = \cdots = X_{k-1} = 0, X_k = 1\}$: $P(S = k) = 2^{-k}$, $P(S = \infty) = 0$, and

$$\mathbb{E}[S] = \sum_{k\ge1} k 2^{-k} = \frac{1}{2} \sum_{k\ge1} k \Big(\frac{1}{2}\Big)^{k-1} = 2, \qquad \text{using } \sum_{k\ge1} k q^{k-1} = \frac{d}{dq} \frac{q}{1-q} = \frac{1}{(1-q)^2}.$$

- (*Doubling strategy*) Doubling a unit stake until the first success gives stake $I_n = 2^{n-1} \mathbb{1}_{\{T > n-1\}}$ (T the waiting time): $\mathbb{E}[I_n] = 2^{n-1} 2^{-(n-1)} = 1$, yet $I_n \to 0$ a.s.

3.2 Infinitely many biased coin tosses

Motivation. Success probability p: moments from the joint law.

With $\Omega = \{0,1\}^{\mathbb{N}}$, X_i and $\mathcal{A}$ as before, there is a unique probability measure $P = P_p$ on $(\Omega, \mathcal{A})$ with

$$P_p(X_{i_1} = x_1, \dots, X_{i_n} = x_n) = p^{\sum_i x_i}(1-p)^{n-\sum_i x_i} \qquad (1 \le i_1 < \cdots < i_n).$$

(Existence for $p = \frac{1}{2}$ was shown above; the general case and uniqueness come in Section 3.6 and Example 3.13.) Then $P_p(X_i = 1) = p$ and $P_p(X_i = X_j = 1) = p^2$ for $i \neq j$, so

$$\mathbb{E}_p[X_i] = p, \qquad \operatorname{Var}(X_i) = p - p^2 = p(1-p), \qquad \operatorname{Cov}(X_i, X_j) = \mathbb{E}_p[X_i X_j] - p^2 = 0 \ (i \neq j).$$

The X_i are pairwise uncorrelated (in fact independent, Example 3.13). For $S_n = X_1 + \cdots + X_n$: $\mathbb{E}_p[S_n] = np$, $\operatorname{Var}(S_n) = np(1-p)$ (Bienaymé).

- $X := \sum_n 2^{-n} X_n$ has $\mathbb{E}_p[X] = \sum_n 2^{-n} p = p$ by Theorem 1.56. The partial sums are Cauchy in L^2, and uncorrelatedness gives $\operatorname{Var}(X) = \sum_n 4^{-n} p(1-p) = \frac{1}{3}p(1-p)$.
- The waiting time S for the first success is **geometric**: $P_p(S = n) = (1-p)^{n-1}p$. Hence $\mathbb{E}_p[S] = \sum_n n(1-p)^{n-1}p = \frac{1}{p}$ and $\operatorname{Var}(S) = \frac{1-p}{p^2}$. [Supplement] For the variance, $\mathbb{E}_p[S(S-1)] = p(1-p)\sum_n n(n-1)(1-p)^{n-2} = \frac{2(1-p)}{p^2}$ (second derivative of the geometric series), so $\operatorname{Var}(S) = \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}$.

3.3 The (strong and the weak) law of large numbers

Motivation. Empirical means of uncorrelated, bounded-variance variables converge in L^2, in probability and a.s.

$X_1, X_2, \cdots \in \mathcal{L}^2$ are uncorrelated with $\sup_i \sigma_i^2 < \infty$, $\sigma_i^2 := \operatorname{Var}(X_i)$, and $S_n := X_1 + \cdots + X_n$ (S_n/n: *empirical mean*).

Remark 3.1

We may assume $\mathbb{E}[X_i] = 0$: the centred $\tilde{X}_i := X_i - \mathbb{E}[X_i]$ are in $\mathcal{L}^2$, uncorrelated, have the same variances, and $\frac{\tilde{S}_n}{n} - \frac{\mathbb{E}[\tilde{S}_n]}{n} = \frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}$.

Theorem 3.2 (L^2 law of large numbers)

$\lim_n \mathbb{E}\Big[\Big(\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\Big)^2\Big] = 0$ (resp. $\mathbb{E}[(\frac{S_n}{n} - m)^2] \to 0$ if $\mathbb{E}[X_i] \equiv m$).

Proof. By Bienaymé (Remark 2.26),

$$\mathbb{E}\Big[\Big(\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\Big)^2\Big] = \operatorname{Var}\Big(\frac{S_n}{n}\Big) = \frac{1}{n^2}\sum_{i=1}^n \sigma_i^2 \le \frac{1}{n}\sup_i \sigma_i^2 \to 0.$$

■

Remark 3.3

This is a Hilbert space fact: in L^2 with $(X,Y)=\mathbb{E}[XY]$, the arithmetic mean of uniformly bounded orthogonal vectors tends to 0, since $\|S_n/n\|^2 = n^{-2}\sum_{i\le n}\|X_i\|^2 \to 0$.

Theorem 3.4 (Weak law of large numbers)

If $X_i \in \mathcal{L}^2$ are uncorrelated with uniformly bounded variances and $\mathbb{E}[X_i]=m$, then for every $\varepsilon>0$, $\lim_n P(|\frac{S_n}{n}-m|\ge\varepsilon)=0$, i.e. $S_n/n \to m$ in probability.

Proof. Chebyshev: $P(|\frac{S_n}{n}-m|\ge\varepsilon) \le \varepsilon^{-2}\operatorname{Var}(\frac{S_n}{n}) \to 0$ by Theorem 3.2. ∎

Example 3.5 (Bernoulli)

For Bernoulli experiments with parameter p, $\mathbb{E}_p[X_i]=p$ and $\operatorname{Var}(X_i)=p(1-p)\le\frac{1}{4}$, so

$$P_p\Big(\Big|\frac{S_n}{n}-p\Big|\ge\varepsilon\Big)\to 0. \tag{3.1}$$

(J. Bernoulli, *Ars Conjectandi*): p is a long-run relative frequency.

Example 3.6 (Bernstein polynomials)

Every $f\in C([0,1])$ is a uniform limit of polynomials. For $p\in[0,1]$ the transformation theorem (Remark 2.15) gives

$$B_n(p) := \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k}p^k(1-p)^{n-k} = \sum_k f\Big(\frac{k}{n}\Big)P_p(S_n=k) = \mathbb{E}_p\Big[f\Big(\frac{S_n}{n}\Big)\Big].$$

With δ from uniform continuity, splitting on $\{|\frac{S_n}{n}-p|\le\delta\}$ and Chebyshev:

$$|B_n(p)-f(p)| \le \mathbb{E}_p\Big|f\Big(\frac{S_n}{n}\Big)-f(p)\Big| \le \varepsilon + 2\|f\|_\infty P_p\Big(\Big|\frac{S_n}{n}-p\Big|>\delta\Big) \le \varepsilon + \frac{2\|f\|_\infty}{4\delta^2 n}.$$

Hence $\limsup_n \|B_n-f\|_\infty \le \varepsilon$ for every ε, i.e. $\|B_n-f\|_\infty\to 0$.

Theorem 3.7 (Strong law of large numbers)

Let $X_1,X_2,\dots\in\mathcal{L}^2$ be uncorrelated with $\sup_i\sigma^2(X_i)=c<\infty$. Then $\frac{S_n}{n}-\frac{\mathbb{E}[S_n]}{n}\to 0$ P-a.s. (resp. $\frac{S_n}{n}\to m$ a.s. if $\mathbb{E}[X_i]\equiv m$).

Proof. By Remark 3.1 let $\mathbb{E}[X_i]=0$.

Step 1 (the subsequence $n_k=k^2$). For $\varepsilon>0$, Chebyshev gives $P(|S_{k^2}|/k^2>\varepsilon)\le\frac{\operatorname{Var}(S_{k^2})}{\varepsilon^2k^4}\le\frac{c}{\varepsilon^2k^2}$, which is summable. By Lemma 2.10, $S_{k^2}/k^2\to 0$ outside a null set N_1.

Step 2 (fluctuations between squares). Let $D_k := \max_{k^2\le l<(k+1)^2}|S_l-S_{k^2}|$. Since $\operatorname{Var}(S_l-S_{k^2})=\sum_{k^2<i\le l}\sigma_i^2\le 2kc$ and there are $2k$ indices l,

$$P\Big(\frac{D_k}{k^2}\ge\varepsilon\Big)\le\sum_{l=k^2+1}^{k^2+2k}P(|S_l-S_{k^2}|\ge\varepsilon k^2)\le 2k\cdot\frac{2kc}{\varepsilon^2k^4}=\frac{4c}{\varepsilon^2k^2}.$$

Lemma 2.10 gives $D_k/k^2\to 0$ outside a null set N_2.

Step 3. For $k^2 \le n < (k+1)^2$: $\frac{|S_n|}{n} \le \frac{|S_{k^2}|+D_k}{k^2} \to 0$ on $(N_1 \cup N_2)^c$. ■

Example 3.8

For Bernoulli experiments, $\frac{1}{n}\sum_{i\le n} X_i \to p$ P_p-a.s. (É. Borel 1909, strengthening Bernoulli's (3.1)). For a fair coin, $S_n = Y_1 + \cdots + Y_n$, $Y_i := 2X_i - 1$, is a **simple random walk** (rescaled: Brownian motion, Section 4.1); $S_n/n \to 0$ a.s., and the exact order is the *law of the iterated logarithm*:

$$\limsup_n \frac{S_n}{\sqrt{2n\log\log n}} = +1, \qquad \liminf_n \frac{S_n}{\sqrt{2n\log\log n}} = -1 \qquad P\text{-a.s.}$$

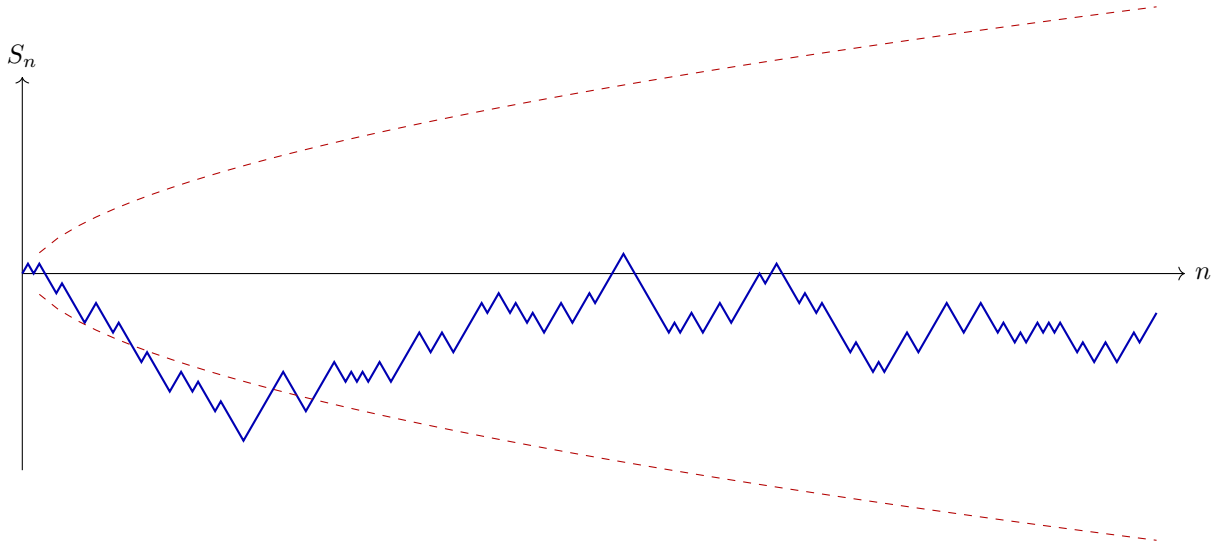


Figure. *A 200-step random walk with $\pm\sqrt{2n\log\log n}$ (dashed).*

3.4 Independence

Motivation. Independence passes from generators to σ-algebras, giving the 0–1 law and Borel–Cantelli 2.

3.4.1 Independent events

Definition 3.9 (Independence)

Events $A_i \in \mathcal{A}$, $i \in I$, are **independent** if $P(\bigcap_{j\in J} A_j) = \prod_{j\in J} P(A_j)$ for every finite $J \subset I$. Families $\mathcal{B}_i \subset \mathcal{A}$, $i \in I$, are **independent** if this holds for all finite $J \subset I$ and all choices $A_j \in \mathcal{B}_j$.

Theorem 3.10

Let $\mathcal{B}_i$, $i \in I$, be independent $\cap$-stable families. Then (i) $\sigma(\mathcal{B}_i)$, $i \in I$, are independent; (ii) for a partition $I = \bigcup_{k\in K} J_k$, the σ-algebras $\sigma(\bigcup_{i\in J_k} \mathcal{B}_i)$, $k \in K$, are independent.

Proof. (i) Let $J = \{j_1, \dots, j_n\}$ and $A_{j_\nu} \in \sigma(\mathcal{B}_{j_\nu})$; we show

$$P(A_{j_1} \cap \cdots \cap A_{j_n}) = P(A_{j_1}) \cdots P(A_{j_n}). \tag{3.2}$$

Step 1. Fix $A_{j_2} \in \mathcal{B}_{j_2}, \dots, A_{j_n} \in \mathcal{B}_{j_n}$. Then $\mathcal{D}_{j_1} := \{A \in \sigma(\mathcal{B}_{j_1}) \mid P(A \cap A_{j_2} \cap \cdots \cap A_{j_n}) = P(A)P(A_{j_2}) \cdots P(A_{j_n})\}$ is a Dynkin system containing $\mathcal{B}_{j_1}$, so $\mathcal{D}_{j_1} = \sigma(\mathcal{B}_{j_1})$ by Theorem 1.8.

Step 2. Now $\sigma(\mathcal{B}_{j_1}), \mathcal{B}_{j_2}, \dots, \mathcal{B}_{j_n}$ are independent $\cap$-stable families; repeat Step 1 for j_2, then j_3, and so on, to get (3.2).

(ii) $\mathcal{C}_k := \{\bigcap_{j \in J} A_j \mid J \subset J_k \text{ finite}, \ A_j \in \mathcal{B}_j\}$ are $\cap$-stable and independent: $P(\bigcap_\nu \bigcap_{i \in J_\nu} A_i) = \prod_\nu P(\bigcap_{i \in J_\nu} A_i)$. By (i) the $\sigma(\mathcal{C}_k)$ are independent, and $\sigma(\mathcal{C}_k) = \sigma(\bigcup_{i \in J_k} \mathcal{B}_i)$. ■

Example 3.11

If A_i, $i \in I$, are independent, then the σ-algebras $\sigma(\{A_i\}) = \{\emptyset, A_i, A_i^c, \Omega\}$ are independent (Theorem 3.10(i) with $\mathcal{B}_i = \{A_i\}$). But $\{A_i, A_i^c \mid i \in I\}$ is not independent unless $P(A_i) \in \{0, 1\}$.

Remark 3.12

Pairwise independence is weaker: for two fair tosses, A = "first 1", B = "second 1", C = "equal" have probability $\frac{1}{2}$, pairwise intersections $\frac{1}{4}$, but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C)$.

Example 3.13 (Independent 0-1 experiments)

On $\Omega = \{0,1\}^{\mathbb{N}}$ with $\mathcal{A} = \sigma(\{X_i = 1\}, i \in \mathbb{N})$ let P_p satisfy (i) $P_p(X_i = 1) = p$ and (ii) the events $\{X_i = 1\}$, $i \in \mathbb{N}$, are independent. By (ii) and Example 3.11, for $i_1 < \cdots < i_n$ and $x_j \in \{0,1\}$

$$P_p(X_{i_1} = x_1, \dots, X_{i_n} = x_n) = \prod_j P_p(X_{i_j} = x_j) = p^k(1-p)^{n-k}, \qquad k = \textstyle\sum_j x_j.$$

These cylinder probabilities form a $\cap$-stable generator, so P_p is uniquely determined by (i) and (ii) (Theorem 1.9).

Theorem 3.14 (Kolmogorov's 0–1 law)

Let $\mathcal{B}_n$, $n \in \mathbb{N}$, be independent σ-algebras and $\mathcal{B}_\infty := \bigcap_{n=1}^\infty \sigma(\bigcup_{m \geq n} \mathcal{B}_m)$ the **tail σ-algebra**. Then $P(A) \in \{0,1\}$ for all $A \in \mathcal{B}_\infty$.

Illustration. For $\mathcal{B}_i = \sigma(\{X_i = 1\})$, $\mathcal{B}_\infty \ni \limsup_i \{X_i = 1\}$, $\{\lim_n \frac{1}{n} \sum_{i \leq n} X_i \text{ exists}\}$.

Proof.

Step 1. By Theorem 3.10(ii), $\mathcal{B}_1, \dots, \mathcal{B}_{n-1}, \sigma(\bigcup_{m \geq n} \mathcal{B}_m) \supset \mathcal{B}_\infty$ are independent for all n, so $\mathcal{B}_\infty, \mathcal{B}_1, \mathcal{B}_2, \dots$ are.

Step 2. Again by Theorem 3.10(ii), $\mathcal{B}_\infty \subset \sigma(\bigcup_n \mathcal{B}_n)$ is independent of $\sigma(\bigcup_n \mathcal{B}_n)$, hence of itself; apply Lemma 3.15. ■

Lemma 3.15

If a σ-algebra $\mathcal{B} \subset \mathcal{A}$ is independent of itself, then $P(A) \in \{0,1\}$ for all $A \in \mathcal{B}$.

Proof. $P(A) = P(A \cap A) = P(A)^2$. ■

For independent A_n, $P(\limsup_n A_n) \in \{0,1\}$ ($\mathcal{B}_n := \sigma(\{A_n\})$, Example 3.11); which value occurs:

Lemma 3.16 (Borel–Cantelli, part 2)

If $A_i \in \mathcal{A}$ are independent and $\sum_i P(A_i) = \infty$, then $P(\limsup_i A_i) = 1$.

Proof. Show $P(\bigcap_{m\geq n} A_m^c) = 0$: by continuity, independence of complements (Example 3.11) and $1-\alpha \leq e^{-\alpha}$,

$$P\Big(\bigcap_{m\geq n} A_m^c\Big) = \lim_k P\Big(\bigcap_{m=n}^{n+k} A_m^c\Big) = \lim_k \prod_{m=n}^{n+k} (1-P(A_m)) \leq \lim_k \exp\Big(-\sum_{m=n}^{n+k} P(A_m)\Big) = 0.$$

■

Example 3.17 (A text appears infinitely often)

Fix a word $(x_1,\dots,x_N)$ with K ones, $p \in (0,1)$; $A_i =$ "the i-th length-N block is the word" are independent (Theorem 3.10(ii)) with probability $\alpha := p^K(1-p)^{N-K} > 0$. Borel–Cantelli 2: infinitely many occurrences a.s.; the strong law: $\frac{1}{n}\sum_{i\leq n} \mathbb{1}_{A_i} \to \alpha$ a.s.

3.4.2 Independent random variables

Definition 3.18 (Independent random variables)

(i) For a r.v. $X : (\Omega,\mathcal{A}) \to (\mathbb{R},\mathcal{B}(\mathbb{R}))$, $\sigma(X) := X^{-1}(\mathcal{B}(\mathbb{R})) = \{\{X \in A\} \mid A \in \mathcal{B}(\mathbb{R})\} \subset \mathcal{A}$ is the σ-algebra **generated by** X.

(ii) R.v.'s X_i, $i \in I$, are **independent** if the $\sigma(X_i)$ are independent, i.e. $P(\bigcap_{j\in J}\{X_j \in A_j\}) = \prod_{j\in J} P(X_j \in A_j)$ for finite J and Borel A_j.

(iii) X_i, $i \in I$, are **independent of** a sub-σ-algebra $\tilde{\mathcal{A}}$ if $\sigma(X_i \mid i \in I)$ (the smallest σ-algebra making all X_i measurable) is independent of $\tilde{\mathcal{A}}$.

Remark 3.19

If X_i, $i \in I$, are independent and $h_i : \mathbb{R} \to \mathbb{R}$ are Borel, then $Y_i := h_i(X_i)$ are independent, because $\sigma(Y_i) \subset \sigma(X_i)$.

Theorem 3.20 (Product rule)

If $X_1,\dots,X_n \geq 0$ are independent, then $\mathbb{E}[X_1\cdots X_n] = \mathbb{E}[X_1]\cdots\mathbb{E}[X_n]$.

Proof.

Step 1 (reduction to $n=2$). $X_1\cdots X_{n-1}$ is $\sigma(\sigma(X_1)\cup\cdots\cup\sigma(X_{n-1}))$-measurable, independent of $\sigma(X_n)$ (Theorem 3.10(ii)); induct. For independent $X, Y \geq 0$ show

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]. \tag{3.3}$$

Step 2 (simple functions). For $X_n = \sum_i \alpha_i \mathbb{1}_{A_i}$, $Y_n = \sum_j \beta_j \mathbb{1}_{B_j}$ with $\alpha_i,\beta_j \geq 0$, $A_i \in \sigma(X)$, $B_j \in \sigma(Y)$:

$$\mathbb{E}[X_nY_n] = \sum_{i,j} \alpha_i\beta_j P(A_i\cap B_j) = \sum_{i,j} \alpha_i\beta_j P(A_i)P(B_j) = \mathbb{E}[X_n]\mathbb{E}[Y_n].$$

Step 3. Choose $\sigma(X)$- resp. $\sigma(Y)$-measurable simple $X_n \nearrow X$, $Y_n \nearrow Y$ (Facts 2.2(v)); then $X_nY_n \nearrow XY$, and the MCT turns Step 2 into (3.3). ■

Corollary 3.21

X, Y independent and in $\mathcal{L}^1 \Rightarrow XY \in \mathcal{L}^1$ and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Proof. $\mathbb{E}[X^{\varepsilon_1}Y^{\varepsilon_2}] = \mathbb{E}[X^{\varepsilon_1}]\mathbb{E}[Y^{\varepsilon_2}] < \infty$ for $\varepsilon_i \in \{+,-\}$ (Remark 3.19); expand $XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+$. ■

Remark 3.22

The converse fails. For $X \sim N(0,1)$ and $Y = X^2$, X, Y are dependent but $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$. Uncorrelatedness is also not preserved under functions: $\mathbb{E}[X^+Y^+] \neq \mathbb{E}[X^+]\mathbb{E}[Y^+]$. [Supplement] Indeed $\mathbb{E}[X^+Y^+] = \int_0^\infty x^3\varphi(x)\,dx = \frac{2}{\sqrt{2\pi}}$ (since $\int_0^\infty x^3 e^{-x^2/2}dx = 2$), while $\mathbb{E}[X^+]\mathbb{E}[Y^+] = \frac{1}{\sqrt{2\pi}} \cdot 1$.
Independent $X, Y \in \mathcal{L}^2$ are uncorrelated.

Corollary 3.23 (Strong law for independent variables)

If $X_1, X_2, \cdots \in \mathcal{L}^2$ are independent with $\sup_i \mathrm{Var}(X_i) < \infty$, then $\frac{1}{n}\sum_{i\le n}(X_i - \mathbb{E}[X_i]) \to 0$ a.s.; if $\mathbb{E}[X_i] \equiv m$, then $\frac{1}{n}\sum_{i\le n} X_i \to m$ a.s.

Proof. Independent $\mathcal{L}^2$ variables are uncorrelated (Remark 3.22); apply Theorem 3.7. ■

3.5 Kolmogorov's law of large numbers

Motivation. For identically distributed, pairwise independent variables a first moment suffices: truncate, use geometric subsequences, interpolate.

(X_i) is **identically distributed** (i.d.) if $P \circ X_i^{-1} = P \circ X_1^{-1}$ for all i; i.i.d. means independent and identically distributed.

Theorem 3.24 (Kolmogorov 1930)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are i.i.d. with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. Independent implies pairwise independent; apply Theorem 3.25. ■

Theorem 3.25 (Etemadi 1981)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are pairwise independent and identically distributed with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. Assume $X_i \ge 0$ (else treat $X_i^{\pm}$).

Step 1 (truncation). $\tilde{X}_i := X_i \mathbb{1}_{\{X_i < i\}}$ are pairwise independent (Remark 3.19); it suffices that $\tilde{S}_n/n \to m$ a.s., $\tilde{S}_n := \sum_{i\le n}\tilde{X}_i$, since

$$\sum_n P(X_n \neq \tilde{X}_n) = \sum_n P(X_1 \ge n) = \sum_n \sum_{k\ge n} P(X_1 \in [k, k+1))$$
$$= \sum_k k\, P(X_1 \in [k, k+1)) \le \mathbb{E}[X_1] < \infty,$$

and by Borel–Cantelli $X_n = \tilde{X}_n$ eventually, a.s.

Step 2 (geometric subsequence). $\alpha > 1$, $k_n := \lfloor \alpha^n \rfloor$; assume (Step 3)

$$\frac{\tilde{S}_{k_n} - \mathbb{E}[\tilde{S}_{k_n}]}{k_n} \to 0 \quad \text{a.s.} \tag{3.4}$$

$\mathbb{E}[\tilde{X}_i] \nearrow m$, so by Cesàro $\tilde{S}_{k_n}/k_n \to m$ off a null N_α. For $k_n \le l < k_{n+1}$, by monotonicity

$$\frac{k_n}{k_{n+1}} \cdot \frac{\tilde{S}_{k_n}}{k_n} \le \frac{\tilde{S}_l}{l} \le \frac{\tilde{S}_{k_{n+1}}}{k_{n+1}} \cdot \frac{k_{n+1}}{k_n},$$

and $k_{n+1}/k_n \to \alpha$, so $\frac{m}{\alpha} \le \liminf_l \frac{\tilde{S}_l}{l} \le \limsup_l \frac{\tilde{S}_l}{l} \le \alpha m$ off N_α; let $\alpha_j \searrow 1$.

Step 3 (proof of (3.4)). By Lemma 2.10 and Chebyshev (pairwise uncorrelated), it is enough that

$$s := \sum_n \frac{1}{k_n^2} \sum_{i \le k_n} \mathbb{E}[\tilde{X}_i^2] = \sum_i \Big(\sum_{n:\, k_n \ge i} \frac{1}{k_n^2} \Big) \mathbb{E}[\tilde{X}_i^2] < \infty.$$

Step 4 (a geometric bound). For some c,

$$\sum_{n:\, k_n \ge i} \frac{1}{k_n^2} \le \frac{c}{i^2}. \tag{3.5}$$

since $k_n > \alpha^n - 1 \ge c_\alpha \alpha^n$, $c_\alpha = \frac{\alpha - 1}{\alpha}$, and the least n_i with $k_n \ge i$ has $\alpha^{n_i} \ge i$:

$$\sum_{n \ge n_i} \frac{1}{k_n^2} \le c_\alpha^{-2} \sum_{n \ge n_i} \alpha^{-2n} = \frac{c_\alpha^{-2} \alpha^{-2n_i}}{1 - \alpha^{-2}} \le \frac{c_\alpha^{-2}}{1 - \alpha^{-2}} \cdot \frac{1}{i^2}.$$

Step 5. With (3.5), $\mathbb{E}[\tilde{X}_i^2] = \mathbb{E}[X_1^2 \mathbb{1}_{\{X_1 < i\}}] \le \sum_{l \le i} l^2 P(X_1 \in [l-1, l))$ and $\sum_{i \ge l} \frac{1}{i^2} \le \frac{1}{l^2} + \sum_{i > l} \big(\frac{1}{i-1} - \frac{1}{i} \big) \le \frac{2}{l}$:

$$\begin{aligned} s &\le c \sum_i \frac{1}{i^2} \sum_{l \le i} l^2 P(X_1 \in [l-1, l)) = c \sum_l l^2 \Big(\sum_{i \ge l} \frac{1}{i^2} \Big) P(X_1 \in [l-1, l)) \\ &\le 2c \sum_l \mathbb{E}\big[l \mathbb{1}_{\{X_1 \in [l-1, l)\}} \big] \le 2c(\mathbb{E}[X_1] + 1). \end{aligned}$$

■

Corollary 3.26

If $X_1, X_2, \cdots \ge 0$ are pairwise independent and identically distributed, then $\frac{1}{n} \sum_{i \le n} X_i \to \mathbb{E}[X_1] \in [0, \infty]$ a.s.

Proof. If $\mathbb{E}[X_1] < \infty$ this is Theorem 3.25. If $\mathbb{E}[X_1] = \infty$, apply Theorem 3.25 to $X_i \wedge N$:

$$\frac{1}{n} \sum_{i \le n} X_i \ge \frac{1}{n} \sum_{i \le n} (X_i \wedge N) \to \mathbb{E}[X_1 \wedge N] \quad \text{a.s.},$$

and $\mathbb{E}[X_1 \wedge N] \nearrow \infty$ (MCT). ■

3.6 Finite products of probability measures

Motivation. The product measure, built by integrating sections, comes with Fubini's theorem.

For probability spaces $(S_1, \mathcal{S}_1, \mu_1)$, $(S_2, \mathcal{S}_2, \mu_2)$, $\Omega := S_1 \times S_2$, $X_i(x_1, x_2) = x_i$, and

$$\mathcal{A} := \mathcal{S}_1 \otimes \mathcal{S}_2 := \sigma(X_1, X_2) = \sigma(\{A_1 \times A_2 \mid A_i \in \mathcal{S}_i\}).$$

we seek the **product measure** $P =: \mu_1 \otimes \mu_2$ with

$$P(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \qquad (A_i \in \mathcal{S}_i). \tag{3.6}$$

unique by Theorem 1.9 (rectangles are a $\cap$-stable generator).

Theorem 3.27 (Fubini for probability measures)

(i) For every $\mathcal{A}$-measurable $f : \Omega \to \overline{\mathbb{R}}$ and $x_1 \in S_1$, $x_2 \in S_2$, the sections $f_{x_1} := f(x_1, \cdot)$ and $f_{x_2} := f(\cdot, x_2)$ are measurable. If moreover $f \geq 0$, then $x_1 \mapsto \int_{S_2} f_{x_1}\, d\mu_2$ and $x_2 \mapsto \int_{S_1} f_{x_2}\, d\mu_1$ are $\mathcal{S}_1$- resp. $\mathcal{S}_2$-measurable.

(ii) There is a unique probability measure $P = \mu_1 \otimes \mu_2$ on $(\Omega, \mathcal{A})$ such that for all $\mathcal{A}$-measurable $f \geq 0$

$$\int_\Omega f\, dP = \int_{S_1} \Big(\int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1), \tag{3.7}$$

and the order may be interchanged:

$$\int_\Omega f\, dP = \int_{S_2} \Big(\int_{S_1} f(x_1, x_2)\, \mu_1(dx_1) \Big) \mu_2(dx_2). \tag{3.8}$$

In particular, with the sections $A_{x_1} := \{x_2 \mid (x_1, x_2) \in A\}$, $A_{x_2} := \{x_1 \mid (x_1, x_2) \in A\}$,

$$P(A) = \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1) = \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2) \qquad (A \in \mathcal{A}). \tag{3.9}$$

Proof.

Step 1 (sections are measurable). $\phi_{x_1}(x_2) := (x_1, x_2)$ is measurable, because $\phi_{x_1}^{-1}(A_1 \times A_2)$ is A_2 or $\emptyset$ and rectangles generate $\mathcal{A}$ (Remark 1.32). So $f_{x_1} = f \circ \phi_{x_1}$ is measurable; likewise f_{x_2}.

Step 2 (indicators). For $f \geq 0$, the map

$$x_1 \mapsto \int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \in [0, \infty] \tag{3.10}$$

is defined; for $f = \mathbb{1}_A$ it is $\mu_2(A_{x_1})$. $\mathcal{D} := \{A \in \mathcal{A} \mid x_1 \mapsto \mu_2(A_{x_1})$ measurable$\}$ is a Dynkin system ($\Omega_{x_1} = S_2$; complements $1 - \mu_2(A_{x_1})$; disjoint unions give series) containing rectangles $(\mathbb{1}_{A_1}(x_1)\mu_2(A_2))$, so $\mathcal{D} = \mathcal{A}$ (Theorem 1.8).

Step 3 (measure-theoretic induction). Linearity for simple $f \geq 0$, then simple $f_n \nearrow f$ (Facts 2.2(v)) and the MCT; likewise in the other order.

Step 4 (existence). $P(A) := \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1)$ has $P(\Omega) = 1$, is σ-additive (Theorem 1.56) and satisfies (3.6). (3.7) holds for indicators, hence for $f \geq 0$ (linearity, MCT).

Step 5 (interchanging the order). $Q(A) := \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2)$ agrees with P on rectangles, so $Q = P$ (Theorem 1.9); extend (3.8) as before; (3.9) is $f = \mathbb{1}_A$. ∎

Remark 3.28

If $f \in \mathcal{L}^1(\mu_1 \otimes \mu_2)$, Theorem 3.27 applies to f^+ and f^-, hence by linearity to f.

Definition 3.29 (Finite products of measurable spaces)

For measurable spaces $(S_i, \mathcal{S}_i)$, $1 \le i \le n$, $\Omega := S_1 \times \cdots \times S_n$ carries the **product σ-algebra**

$$\mathcal{A} := \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n = \bigotimes_{i=1}^{n} \mathcal{S}_i := \sigma(\{A_1 \times \cdots \times A_n \mid A_i \in \mathcal{S}_i\}).$$

Equivalently $\mathcal{A} = \sigma(X_1, \dots, X_n)$ with the coordinate projections X_i. [Supplement] *Reason:* $A_1 \times \cdots \times A_n = \bigcap_i X_i^{-1}(A_i)$, and $X_i^{-1}(A_i)$ is a rectangle.

Theorem 3.30 (Finite products of probability measures)

For probability spaces $(S_i, \mathcal{S}_i, \mu_i)$, $1 \le i \le n$, there is a unique probability measure $P = \mu_1 \otimes \cdots \otimes \mu_n = \bigotimes_{i=1}^{n} \mu_i$ on $(S_1 \times \cdots \times S_n, \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n)$ with $P(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$.

Proof. Uniqueness: Theorem 1.9. Existence by induction: $P := P_{n-1} \otimes \mu_n$ (Theorem 3.27), as $\mathcal{A}_{n-1} \otimes \mathcal{S}_n = \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$ (same generating rectangles), and $P(A_1 \times \cdots \times A_n) = P_{n-1}(A_1 \times \cdots \times A_{n-1})\mu_n(A_n) = \prod_i \mu_i(A_i)$. ∎

Theorem 3.31 (Fubini for finite products)

With $P = \mu_1 \otimes \cdots \otimes \mu_n$ and $\mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$-measurable $f \ge 0$,

$$\int f \, dP = \int_{S_1} \cdots \int_{S_n} f(x_1, \dots, x_n) \, \mu_n(dx_n) \cdots \mu_1(dx_1), \tag{3.11}$$

and for every permutation π of $\{1, \dots, n\}$

$$\int f \, dP = \int_{S_{\pi(1)}} \cdots \int_{S_{\pi(n)}} f(x_1, \dots, x_n) \, \mu_{\pi(n)}(dx_{\pi(n)}) \cdots \mu_{\pi(1)}(dx_{\pi(1)}). \tag{3.12}$$

Proof.

Step 1 ((3.11)). Induction: $\int f \, dP = \int_{\Omega_{n-1}} \left(\int_{S_n} f \, d\mu_n \right) dP_{n-1}$ (Theorem 3.27), and the inner integral is $\mathcal{A}_{n-1}$-measurable and ≥ 0.

Step 2 ((3.12)) [Supplement]. Regrouping $\Omega = \left(\prod_{i \ne \pi(n)} S_i \right) \times S_{\pi(n)}$ keeps the product σ-algebra, so the iterated integrals are defined (Theorem 3.27(i), induction). For $f = \mathbb{1}_A$ the right side is a probability measure in A agreeing with P on rectangles, hence $= P$ (Theorem 1.9); extend by linearity and the MCT. ∎

Remark 3.32

If $f \in \mathcal{L}^1(\mu_1 \otimes \cdots \otimes \mu_n)$, Theorem 3.31 applies to $f^{\pm}$ and hence to f.

3.7 Weak convergence of probability measures

Motivation. Distributions converge when integrals of bounded continuous functions do ($\delta_{1/n} \not\to \delta_0$ setwise).

Definition 3.33 (Weak convergence)

Let Ω be a metric space with Borel σ-algebra $\mathcal{B}(\Omega)$, and μ, μ_n probability measures on it. $\mu_n \to \mu$ **weakly** if $\int f\,d\mu_n \to \int f\,d\mu$ for all $f \in C_b(\Omega)$ (bounded continuous functions).

Example 3.34

(i) $x_n \to x$ implies $\delta_{x_n} \to \delta_x$ weakly, since $f(x_n) \to f(x)$ for continuous f. (ii) $\mu_n := N(0, \frac{1}{n}) \to \delta_0$ weakly: substituting $x = y/\sqrt{n}$ and using the DCT,

$$\int f\,d\mu_n = \int f(x)\frac{e^{-nx^2/2}}{\sqrt{2\pi/n}}\,dx = \int f\Big(\frac{y}{\sqrt{n}}\Big)\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy \to f(0) = \int f\,d\delta_0.$$

(Also for finite measures.)

Theorem 3.35 (Portmanteau theorem)

For probability measures μ, μ_n on a metric space (Ω, d) the following are equivalent:

(i) $\mu_n \to \mu$ weakly;

(ii) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded uniformly continuous f;

(iii) $\limsup_n \mu_n(F) \le \mu(F)$ for all closed F;

(iv) $\liminf_n \mu_n(G) \ge \mu(G)$ for all open G;

(v) $\mu_n(A) \to \mu(A)$ for all μ-continuity sets A, i.e. $A \in \mathcal{B}(\Omega)$ with $\mu(\partial A) = 0$;

(vi) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded measurable f that are continuous μ-a.e.

Proof.

Step Easy implications. (i)⇒(ii) and (vi)⇒(i) are special cases; (iii)⇔(iv) by taking complements.

Step (ii)⇒(iii). Open $G_m := \{d(\cdot, F) < \frac{1}{m}\} \searrow F$, so $\mu(G_m) < \mu(F) + \varepsilon$ for some m. With $\phi = 1$ on $(-\infty, 0]$, $1 - x$ on $[0, 1]$, 0 on $[1, \infty)$, $f := \phi(m\,d(\cdot, F))$ is uniformly continuous, $\mathbb{1}_F \le f \le \mathbb{1}_{G_m}$, so $\limsup_n \mu_n(F) \le \lim_n \int f\,d\mu_n = \int f\,d\mu \le \mu(G_m) < \mu(F) + \varepsilon$.

Step (iii)⇒(v). With interior $\mathring{A}$ and closure $\bar{A}$, $\mu(\partial A) = 0$ gives $\mu(\mathring{A}) = \mu(A) = \mu(\bar{A})$, and

$$\mu(A) = \mu(\mathring{A}) \overset{\text{(iv)}}{\le} \liminf_n \mu_n(\mathring{A}) \le \liminf_n \mu_n(A) \le \limsup_n \mu_n(A) \le \limsup_n \mu_n(\bar{A}) \overset{\text{(iii)}}{\le} \mu(\bar{A}) = \mu(A).$$

Step (v)⇒(vi). $D := \{x \mid \mu(f = x) \ne 0\}$ is countable (Remark 2.16); pick $c_0 < \cdots < c_m$ off D with $c_{k+1} - c_k \le \varepsilon$, $c_0 \le f < c_m$. $A_k := f^{-1}([c_k, c_{k+1}))$ are μ-continuity sets, since (below)

$$\partial A_k \subset \{f = c_k\} \cup \{f = c_{k+1}\} \cup D_f, \qquad D_f := \{\omega \mid f \text{ is discontinuous at } \omega\}, \tag{3.13}$$

and all three sets are μ-null. With $g := \sum_k c_k \mathbb{1}_{A_k}$, $\|f - g\|_\infty \le \varepsilon$ and

$$\begin{aligned}\Big|\int f\,d\mu - \int f\,d\mu_n\Big| &\le \int |f - g|\,d\mu + \Big|\int g\,d\mu - \int g\,d\mu_n\Big| + \int |g - f|\,d\mu_n \\ &\le 2\varepsilon + \sum_k |c_k||\mu(A_k) - \mu_n(A_k)| \to 2\varepsilon.\end{aligned}$$

Step Proof of (3.13). For $\omega \in \partial A_k \setminus D_f$ and $\varepsilon' > 0$, some $B(\omega, \delta)$ maps into $B(f(\omega), \varepsilon')$ and meets A_k and A_k^c, so $B(f(\omega), \varepsilon')$ meets $[c_k, c_{k+1})$ and its complement: $f(\omega) \in \{c_k, c_{k+1}\}$.

[Supplement] $D_f = \bigcup_k \{\mathrm{osc}_f \ge \frac{1}{k}\}$, $\mathrm{osc}_f(\omega) := \lim_{\delta \to 0} \sup\{|f(x) - f(y)| \mid x, y \in B(\omega, \delta)\}$, is measurable: each set is closed (a small ball with oscillation $< \frac{1}{k}$ serves nearby points). ■

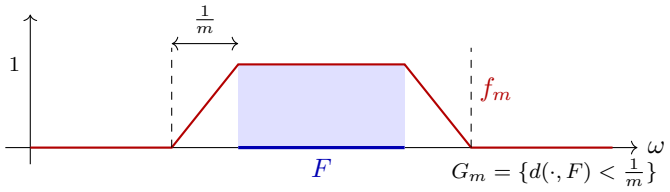


Figure. *The test function of (ii)⇒(iii):* $\mathbb{1}_F \le f_m \le \mathbb{1}_{G_m}$.

Definition 3.36 (Support, C_0)

For continuous f on a metric space Ω, $\operatorname{supp} f := \overline{\{f \neq 0\}}$, and $C_0(\Omega) := \{f \in C_b(\Omega) \mid \operatorname{supp} f \text{ compact}\}$.

Corollary 3.37

For probability measures μ, μ_n on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with distribution functions F, F_n, the following are equivalent:

(i) $\mu_n \to \mu$ **vaguely**: $\int f\, d\mu_n \to \int f\, d\mu$ for all $f \in C_0(\mathbb{R})$;

(ii) $\mu_n \to \mu$ weakly;

(iii) $F_n(x) \to F(x)$ at every continuity point x of F;

(iv) $\mu_n((a,b]) \to \mu((a,b])$ whenever $\mu(\{a\}) = \mu(\{b\}) = 0$.

Proof.

Step (i)⇒(ii) [Supplement]. For $f \in C_b(\mathbb{R})$ take $\mu([-R,R]) > 1-\varepsilon$ and $g \in C_0(\mathbb{R})$, $0 \le g \le 1$, $g = 1$ on $[-R,R]$; then $\int (1-g)\, d\mu_n < \varepsilon$ eventually, and with $fg \in C_0(\mathbb{R})$

$$\Big|\int f\, d\mu_n - \int f\, d\mu\Big| \le \Big|\int fg\, d\mu_n - \int fg\, d\mu\Big| + \|f\|_\infty \Big(\int (1-g)\, d\mu_n + \int (1-g)\, d\mu\Big),$$

whose lim sup is at most $2\|f\|_\infty \varepsilon$.

Step (ii)⇒(iii). If F is continuous at x, then $\mu(\{x\}) = 0 = \mu(\partial(-\infty, x])$, and Portmanteau (v) gives $F_n(x) = \mu_n((-\infty, x]) \to F(x)$.

Step (iii)⇒(iv). F is continuous at a and b, so $\mu((a,b]) = F(b) - F(a) = \lim_n (F_n(b) - F_n(a))$.

Step (iv)⇒(i). $D := \{x \mid \mu(\{x\}) \neq 0\}$ is countable. For $f \in C_0(\mathbb{R})$ (uniformly continuous, δ for ε) pick $c_0 < \cdots < c_m$ off D, $\operatorname{supp} f \subset (c_0, c_m]$, $c_k - c_{k-1} \le \delta$; $g := \sum_{k=1}^m f(c_{k-1}) \mathbb{1}_{(c_{k-1}, c_k]}$ has $\|f - g\|_\infty \le \varepsilon$ and

$$\Big|\int f\, d\mu - \int f\, d\mu_n\Big| \le 2\varepsilon + \sum_k |f(c_{k-1})| |\mu((c_{k-1}, c_k]) - \mu_n((c_{k-1}, c_k])| \to 2\varepsilon.$$

■

3.8 Joint distribution and convolution

Motivation. $S_n/n \to \delta_{\mathbb{E}[X_1]}$ weakly; the interesting scale $S_n/\sqrt{n}$ needs laws of sums, i.e. convolutions.

Definition 3.38 (Joint distribution)

For real r.v.'s $X_1, \dots, X_n$ on $(\Omega, \mathcal{A}, P)$, the law $P_{\bar{X}} := P \circ \bar{X}^{-1}$ of $\bar{X} := (X_1, \dots, X_n) : \Omega \to \mathbb{R}^n$ is the **joint distribution** of $X_1, \dots, X_n$.

Remark 3.39

(i) $P_{\bar{X}}(A) = P(\bar{X} \in A)$, $A \in \mathcal{B}(\mathbb{R}^n)$, is a probability measure (Proposition 2.13), because $\bar{X}$ is $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable. Indeed

$$\mathcal{B}(\mathbb{R}^n) = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R}) = \sigma(\{A_1 \times \cdots \times A_n \mid A_i = (-\infty, x_i]\}),$$

and $\bar{X}^{-1}(A_1 \times \cdots \times A_n) = \bigcap_i \{X_i \in A_i\} \in \mathcal{A}$ (Remark 1.32). [Supplement] *The equalities:* open sets are countable unions of open boxes (Remark 1.27(i)), and half-open boxes are finite differences of orthants.

(ii) By Theorem 1.9, $P_{\bar{X}}$ is determined by $P_{\bar{X}}(A_1 \times \cdots \times A_n) = P(X_1 \in A_1, \dots, X_n \in A_n)$, even with $A_i = (-\infty, x_i]$.

Remark 3.40 (Transformation theorem)

For Borel $\Phi : \mathbb{R}^n \to \mathbb{R}$ with $\Phi(\bar{X}) \in \mathcal{L}^1$ or $\Phi \geq 0$: $\mathbb{E}[\Phi(\bar{X})] = \int_\Omega \Phi(\bar{X})\, dP = \int_{\mathbb{R}^n} \Phi\, dP_{\bar{X}}$ (same proof as in Remark 2.15).

Example 3.41

(i) X, Y uniform on $[0,1]$: independent gives uniform on $[0,1]^2$ (Theorem 3.42); $X = Y$ gives the image of $\lambda|_{[0,1]}$ under $x \mapsto (x, x)$. (ii) Independent $X, Y \sim N(m, \sigma^2)$: joint density $f(x,y) = \frac{1}{2\pi\sigma^2} \exp\big(-\frac{(x-m)^2+(y-m)^2}{2\sigma^2}\big)$.

Theorem 3.42

Let $X_1, \dots, X_n$ have laws $\mu_1, \dots, \mu_n$ and joint law $P_{\bar{X}}$. Then $X_1, \dots, X_n$ are independent $\iff$ $P_{\bar{X}} = \bigotimes_i \mu_i$, i.e. $P_{\bar{X}}(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$. In this case:

(i) $P_{\bar{X}}$ is determined by $\mu_1, \dots, \mu_n$;

(ii) for every permutation π and $\phi \geq 0$ (or $P_{\bar{X}}$-integrable), $\int \phi\, dP_{\bar{X}}$ equals the iterated integral in the order π;

(iii) if $\mu_i(dx_i) = f_i(x_i)\, dx_i$, then $P_{\bar{X}}(d\bar{x}) = \bar{f}(\bar{x})\, d\bar{x}$ with $\bar{f}(\bar{x}) = \prod_i f_i(x_i)$.

Proof. The equivalence is Definition 3.18 rewritten, since $\mu_i(A_i) = P(X_i \in A_i)$.

(i) Theorem 3.30.

(ii) Theorem 3.31 and Remark 3.32.

(iii) $\int \bar{f}\, d\bar{x} = \prod_i \int f_i = 1$ (Fubini; Lebesgue measure on $\mathbb{R}^n$ is the product measure by Theorem 1.9), and $\bar{\mu}(A) := \int_A \bar{f}\, d\bar{x}$ agrees with $P_{\bar{X}}$ on rectangles, so $\bar{\mu} = P_{\bar{X}}$ (Theorem 1.9). ■

Law of S_n for independent summands, with $T_x(y) := x + y$:

Theorem 3.43 (Convolution)

Let X_1, X_2 be independent with laws μ_1, μ_2.

(i) $P_{X_1+X_2} = \mu_1 * \mu_2 := \int \mu_2 \circ T_{x_1}^{-1}(\cdot)\, \mu_1(dx_1)$, i.e. for $A \in \mathcal{B}(\mathbb{R})$

$$\mu_1 * \mu_2(A) = \iint \mathbb{1}_A(x_1 + x_2)\, \mu_1(dx_1)\, \mu_2(dx_2) = \int \mu_2(A - x_1)\, \mu_1(dx_1),$$

where $A - x_1 := \{x - x_1 \mid x \in A\}$.

(ii) If μ_2 has density f_2, then $\mu_1 * \mu_2$ has density $f(x) = \int f_2(x - x_1)\, \mu_1(dx_1)$, which equals $(f_1 * f_2)(x) := \int f_1(x_1) f_2(x - x_1)\, dx_1$ if $\mu_1(dx_1) = f_1(x_1)\, dx_1$.

Proof. (i) With $\bar{A} := \{(x_1, x_2) \mid x_1 + x_2 \in A\} \in \mathcal{B}(\mathbb{R}^2)$ and Theorem 3.42,

$$P(X_1 + X_2 \in A) = (\mu_1 \otimes \mu_2)(\bar{A}) = \int \Big(\int \mathbb{1}_{A - x_1}(x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1) = \int \mu_2(A - x_1)\, \mu_1(dx_1).$$

(ii) Substituting $x = x_1 + x_2$ and using Fubini (Theorem 3.42(ii)):

$$\begin{aligned}(\mu_1 * \mu_2)(A) &= \int \int_{A - x_1} f_2(x_2)\, dx_2\, \mu_1(dx_1) = \int \int_A f_2(x - x_1)\, dx\, \mu_1(dx_1) \\ &= \int_A \Big(\int f_2(x - x_1)\, \mu_1(dx_1) \Big) dx.\end{aligned}$$

■

Example 3.44

(i) If $X_i \sim N(m_i, \sigma_i^2)$ are independent, then $X_1 + X_2 \sim N(m_1 + m_2, \sigma_1^2 + \sigma_2^2)$, i.e. $f_{m_1,\sigma_1^2} * f_{m_2,\sigma_2^2} = f_{m_1+m_2,\sigma_1^2+\sigma_2^2}$. [Supplement] After shifting, let $m_i = 0$, $a = \sigma_1^2$, $b = \sigma_2^2$. Completing the square in y,

$$\int e^{-\frac{y^2}{2a} - \frac{(x-y)^2}{2b}}\, dy = e^{-\frac{x^2}{2(a+b)}} \int e^{-\frac{a+b}{2ab}(y - \frac{ax}{a+b})^2}\, dy = e^{-\frac{x^2}{2(a+b)}} \sqrt{\frac{2\pi ab}{a+b}},$$

and multiplying by $\frac{1}{2\pi\sqrt{ab}}$ gives $f_{0,a+b}(x)$.

(ii) The **Gamma distribution** $\Gamma_{\alpha,p}$ has density $\gamma_{\alpha,p}(x) = \frac{\alpha^p}{\Gamma(p)} x^{p-1} e^{-\alpha x}$ for $x > 0$ (0 otherwise). If $X_i \sim \Gamma_{\alpha,p_i}$ are independent, then $X_1 + X_2 \sim \Gamma_{\alpha,p_1+p_2}$. [Supplement] For $x > 0$, substituting $y = xt$,

$$\begin{aligned}(\gamma_{\alpha,p_1} * \gamma_{\alpha,p_2})(x) &= \frac{\alpha^{p_1+p_2} e^{-\alpha x}}{\Gamma(p_1)\Gamma(p_2)} \int_0^x y^{p_1-1}(x - y)^{p_2-1} dy \\ &= \frac{\alpha^{p_1+p_2} e^{-\alpha x} x^{p_1+p_2-1}}{\Gamma(p_1)\Gamma(p_2)} B(p_1, p_2) = \gamma_{\alpha,p_1+p_2}(x),\end{aligned}$$

using $B(p_1, p_2) = \Gamma(p_1)\Gamma(p_2)/\Gamma(p_1 + p_2)$. For $p_i = 1$: a sum $S_n = T_1 + \cdots + T_n$ of independent $\mathrm{Exp}(\alpha)$ variables has the Gamma density $\gamma_{\alpha,n}(x) = \frac{\alpha^n}{(n-1)!} x^{n-1} e^{-\alpha x}$, $x > 0$.

Example 3.45 (The waiting time paradox)

Independent $\mathrm{Exp}(\alpha)$ gaps T_i ($\mathbb{E}[T_i] = \frac{1}{\alpha}$, e.g. buses), $S_n := T_1 + \cdots + T_n$, $S_0 := 0$; for fixed $t > 0$, $X = t - S_n$, $Y = S_{n+1} - t$ on $\{S_n \le t < S_{n+1}\}$. Not $\mathbb{E}[Y] = \frac{1}{2\alpha}$, but:

(i) $Y \sim \mathrm{Exp}(\alpha)$, so $\mathbb{E}[Y] = \frac{1}{\alpha}$;

(ii) X is Exp(α) "compressed to $[0,t]$": $P(X \geq s) = e^{-\alpha s}$ for $0 \leq s \leq t$ and $P(X = t) = e^{-\alpha t}$, so $\mathbb{E}[X] = \int_0^t s\alpha e^{-\alpha s} ds + te^{-\alpha t} = \frac{1}{\alpha}(1 - e^{-\alpha t})$;

(iii) X and Y are independent.

So $[t - X, t + Y]$ has mean length $\frac{1}{\alpha}(2 - e^{-\alpha t}) \approx \frac{2}{\alpha}$: t tends to fall in long gaps.

Proof. Split by the number n of events in $[0,t]$ (independent $S_n \sim \gamma_{\alpha,n}$, T_{n+1}); for $0 \leq x \leq t$, $y \geq 0$:

$$\begin{aligned} P(X \geq x, Y \geq y) &= P(T_1 \geq t + y) + \sum_{n \geq 1} P(S_n \leq t - x,\ T_{n+1} \geq t + y - S_n) \\ &= e^{-\alpha(t+y)} + \sum_{n \geq 1} \int_0^{t-x} \gamma_{\alpha,n}(s) e^{-\alpha(t+y-s)}\, ds \\ &= e^{-\alpha(t+y)} \Big(1 + \int_0^{t-x} e^{\alpha s} \sum_{n \geq 1} \gamma_{\alpha,n}(s)\, ds\Big). \end{aligned}$$

Since $\sum_{n \geq 1} \gamma_{\alpha,n}(s) = \alpha e^{-\alpha s} \sum_{n \geq 1} \frac{(\alpha s)^{n-1}}{(n-1)!} = \alpha$, the bracket is $1 + (e^{\alpha(t-x)} - 1) = e^{\alpha(t-x)}$, and $P(X \geq x, Y \geq y) = e^{-\alpha y} e^{-\alpha x}$. Setting $x = 0$ gives (i), $y = 0$ gives (ii), and the product form gives (iii).

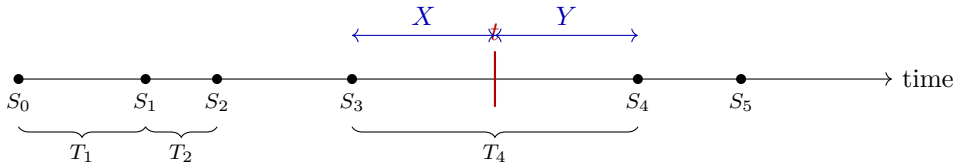


Figure. *The gap containing t (here T_4) is size-biased.*

3.9 Characteristic functions

Motivation. Fourier transforms of laws turn convolutions into products and weak convergence into pointwise convergence.

$\mathcal{M}_+^1(\mathbb{R}^n)$: probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. The **characteristic function** of μ is

$$\hat{\mu}(u) := \int e^{i\langle u,y\rangle} \mu(dy) := \int \cos\langle u, y\rangle\, \mu(dy) + i \int \sin\langle u, y\rangle\, \mu(dy), \qquad u \in \mathbb{R}^n.$$

Theorem 3.46

For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$: (i) $|\hat{\mu}(u)| \leq \hat{\mu}(0) = 1$; (ii) $\hat{\mu}(-u) = \overline{\hat{\mu}(u)}$; (iii) $\hat{\mu}$ is uniformly continuous; (iv) $\hat{\mu}$ is positive definite: $\sum_{j,k=1}^m c_j \bar{c}_k \hat{\mu}(u_j - u_k) \geq 0$ for $c_j \in \mathbb{C}$, $u_j \in \mathbb{R}^n$.

Proof [Supplement]. (i) $|\hat{\mu}(u)| \leq \int |e^{i\langle u,y\rangle}|\, d\mu = 1 = \hat{\mu}(0)$.

(ii) Conjugate the integrand.

(iii) $|\hat{\mu}(u + h) - \hat{\mu}(u)| \leq \int |e^{i\langle h,y\rangle} - 1|\, \mu(dy) \to 0$ (DCT), uniformly in u.

(iv) $\sum_{j,k} c_j \bar{c}_k \hat{\mu}(u_j - u_k) = \int \big|\sum_j c_j e^{i\langle u_j,y\rangle}\big|^2 \mu(dy) \geq 0$. ∎

Theorem 3.47 (Uniqueness theorem)

If $\mu_1, \mu_2 \in \mathcal{M}_+^1(\mathbb{R}^n)$ and $\hat{\mu}_1 = \hat{\mu}_2$, then $\mu_1 = \mu_2$.

Proof [Supplement] (Gaussian smoothing).

Step 1 (a Fourier formula for the Gaussian). For $p_\sigma(x) := (2\pi\sigma^2)^{-n/2} e^{-\|x\|^2/(2\sigma^2)}$, Theorem 3.52 (independent of this section) with $y = \sigma u$, $u \to x/\sigma$ gives

$$p_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2}\, du.$$

Step 2 (smoothed measure is determined by $\hat{\mu}$). $h_\sigma(x) := \int p_\sigma(x-y)\,\mu(dy)$; by Step 1 and Fubini (bound $e^{-\sigma^2\|u\|^2/2}$),

$$h_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \Big(\int e^{-i\langle y,u\rangle} \mu(dy) \Big) du = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \hat{\mu}(-u)\, du,$$

which depends on μ only through $\hat{\mu}$.

Step 3 (letting $\sigma \to 0$). For $f \in C_b(\mathbb{R}^n)$, with γ the standard Gaussian on $\mathbb{R}^n$ and the substitution $x = y + \sigma z$,

$$\int f(x) h_\sigma(x)\, dx = \iint f(y+\sigma z)\, \gamma(dz)\, \mu(dy) \xrightarrow[\sigma\to 0]{} \int f\, d\mu$$

(DCT); the left side is the same for μ_1, μ_2 (Step 2), so $\mu_1 = \mu_2$ (Example 2.30). ∎

Theorem 3.48 (Bochner)

If $\phi : \mathbb{R}^n \to \mathbb{C}$ is continuous, positive definite and $\phi(0) = 1$, there is exactly one $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat{\mu} = \phi$.

On the proof

Uniqueness is Theorem 3.47. Existence is not proved in the lectures; see Klenke, *Probability Theory*, Thm. 15.29. It is not used below.

Theorem 3.49 (Lévy's continuity theorem)

Let $\mu_m \in \mathcal{M}_+^1(\mathbb{R}^n)$.

(i) If $\mu_m \to \mu$ weakly, then $\hat{\mu}_m \to \hat{\mu}$ uniformly on compact sets.

(ii) If $\hat{\mu}_m \to \phi$ pointwise for some $\phi : \mathbb{R}^n \to \mathbb{C}$ continuous at 0, then there is a unique $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat{\mu} = \phi$, and $\mu_m \to \mu$ weakly.

Proof [Supplement] (part (ii) for $n = 1$, the case used in the CLT).

Step (i) Tightness. $\mu(\{\|y\| \ge R\}) < \varepsilon$ and Portmanteau (iii) give $\mu_m(\{\|y\| \ge R\}) < \varepsilon$ for $m \ge m_0$; enlarge R for $m < m_0$.

Step (i) Equicontinuity. $|\hat{\mu}_m(u) - \hat{\mu}_m(v)| \le \int_{\{\|y\|<R\}} |e^{i\langle u-v,y\rangle} - 1|\, d\mu_m + 2\varepsilon \le R\|u-v\| + 2\varepsilon$; with pointwise convergence ($\cos, \sin \in C_b$) and a finite cover, convergence is uniform on compacts.

Step (ii) Tightness from ϕ. For $\delta > 0$ and $\nu \in \mathcal{M}_+^1(\mathbb{R})$, Fubini gives

$$\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \hat{\nu}(u))\, du = 2 \int \Big(1 - \frac{\sin \delta x}{\delta x}\Big) \nu(dx) \ge \nu(\{|x| \ge 2/\delta\}),$$

as $1-\frac{\sin s}{s}\geq\frac{1}{2}$ for $|s|\geq 2$. Continuity of ϕ at 0, $\phi(0)=1$, gives δ with $\frac{1}{\delta}\int_{-\delta}^{\delta}|1-\phi|<\varepsilon$, and by the DCT $\mu_m(\{|x|\geq 2/\delta\})<2\varepsilon$ for large m: tightness.

Step (ii) Helly selection. A diagonal subsequence has $F_{m_k}(q)\to H(q)$ for rational q; $G(x):=\inf_{q>x}H(q)$ is increasing, right continuous, and at continuity points $H(q)\leq\liminf_k F_{m_k}(x)\leq\limsup_k F_{m_k}(x)\leq H(q')$ $(q<x<q')$ gives $F_{m_k}(x)\to G(x)$. Tightness gives $G(\mp\infty)=0,1$, so G is the distribution function of some ν (Remark 2.15) and $\mu_{m_k}\to\nu$ (Corollary 3.37).

Step (ii) Identification. $\hat{\nu}=\phi$ by (i). Every subsequence has a further one converging to some ν' with $\hat{\nu}'=\phi$, so $\nu'=\nu=:\mu$ (Theorem 3.47); a subsequence with $|\int f\,d\mu_m-\int f\,d\mu|\geq\varepsilon$ would contradict this.

($n\geq 2$: Helly with multivariate distribution functions; not needed.) ■

For an $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable $\bar{X}=(X_1,\ldots,X_n)$ on $(\Omega,\mathcal{A},P)$,

$$\varphi_{\bar{X}}(u):=\hat{P}_{\bar{X}}(u)=\int e^{i\langle u,y\rangle}P_{\bar{X}}(dy)=\mathbb{E}\big[e^{i\langle u,\bar{X}\rangle}\big]$$

is the **characteristic function of** $\bar{X}$.

Remark 3.50

$X_1,\ldots,X_n$ are independent iff $\varphi_{\bar{X}}(u_1,\ldots,u_n)=\prod_j\varphi_{X_j}(u_j)$ for all $u\in\mathbb{R}^n$. "⇒": Corollary 3.21 on real and imaginary parts. "⇐": the right side is the characteristic function of $\bigotimes_j P_{X_j}$ (Fubini); apply Theorems 3.47, 3.42.

Theorem 3.51

If $X_1,\ldots,X_n$ are independent, $\alpha\in\mathbb{R}$ and $S:=\alpha\sum_k X_k$, then $\varphi_S(u)=\prod_{k=1}^n\varphi_{X_k}(\alpha u)$.

Proof. $\varphi_S(u)=\mathbb{E}\big[\prod_k e^{i\alpha u X_k}\big]=\prod_k\mathbb{E}[e^{i\alpha u X_k}]$ by independence (Corollary 3.21, applied to the bounded complex factors via real and imaginary parts and induction). ■

Theorem 3.52 (Gaussian characteristic function)

For all $u\in\mathbb{R}^n$: $(2\pi)^{-n/2}\int_{\mathbb{R}^n}e^{i\langle u,y\rangle}e^{-\frac{1}{2}\|y\|^2}dy=e^{-\frac{1}{2}\|u\|^2}$.

Proof [Supplement]. By Fubini the integral factorises over coordinates, so let $n=1$ and $\varphi(u):=(2\pi)^{-1/2}\int e^{iuy}e^{-y^2/2}dy$. Differentiating under the integral (dominated by $|y|e^{-y^2/2}$) and integrating by parts,

$$\varphi'(u)=(2\pi)^{-1/2}\int iy\,e^{iuy}e^{-y^2/2}dy=(2\pi)^{-1/2}\int i\,e^{iuy}\,\frac{d}{dy}\big(-e^{-y^2/2}\big)dy=-u\varphi(u).$$

With $\varphi(0)=1$ this gives $\varphi(u)=e^{-u^2/2}$. ■

Example 3.53

(i) $\hat{\delta}_a(u)=e^{iua}$. (ii) For $\mu=\sum_i\alpha_i\delta_{a_i}$ $(\alpha_i\geq 0,\ \sum\alpha_i=1)$, $\hat{\mu}(u)=\sum_i\alpha_i e^{iua_i}$. In particular the binomial distribution $\beta_n^p=\sum_k\binom{n}{k}p^kq^{n-k}\delta_k$ $(q=1-p)$ has $\hat{\beta}_n^p(u)=(q+pe^{iu})^n$, and the Poisson distribution $\pi_\alpha=\sum_n e^{-\alpha}\frac{\alpha^n}{n!}\delta_n$ has $\hat{\pi}_\alpha(u)=e^{-\alpha}\sum_n\frac{(\alpha e^{iu})^n}{n!}=e^{\alpha(e^{iu}-1)}$.

3.10 Central limit theorem

Motivation. Standardised sums of many independent, individually negligible (Lindeberg) terms are approximately normal.

Definition 3.54 (Central limit property)

Let $X_1, X_2, \dots \in \mathcal{L}^2$ be independent with positive variances, $S_n := X_1 + \cdots + X_n$ and $S_n^* := \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\mathrm{Var}(S_n)}}$ (so $\mathbb{E}[S_n^*] = 0$, $\mathrm{Var}(S_n^*) = 1$). (X_n) has the **central limit property** (CLP) if $P_{S_n^*} \to N(0,1)$ weakly, equivalently (Corollary 3.37)

$$\lim_n P(S_n^* \le b) = \Phi(b) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{b} e^{-x^2/2} dx \qquad \forall b \in \mathbb{R}.$$

Theorem 3.55 (Central limit theorem, Lindeberg)

Let $X_1, X_2, \dots \in \mathcal{L}^2$ be independent, $\sigma_n^2 := \mathrm{Var}(X_n) > 0$, $s_n := (\sum_{k \le n} \sigma_k^2)^{1/2}$, and assume **Lindeberg's condition**

$$L_n(\varepsilon) := \sum_{k=1}^{n} \int_{\{|X_k - \mathbb{E}[X_k]| \ge \varepsilon s_n\}} \Big(\frac{X_k - \mathbb{E}[X_k]}{s_n} \Big)^2 dP \xrightarrow[n\to\infty]{} 0 \qquad \forall \varepsilon > 0. \tag{L}$$

Then (X_n) has the CLP.

Remark 3.56

(i) i.i.d. sequences in $\mathcal{L}^2$ satisfy (L): with $m = \mathbb{E}[X_1]$, $\sigma^2 = \mathrm{Var}(X_1)$, $s_n^2 = n\sigma^2$ and $L_n(\varepsilon) = \sigma^{-2} \int_{\{|X_1 - m| \ge \varepsilon \sqrt{n}\sigma\}} (X_1 - m)^2 dP \to 0$ by the DCT.

(ii) **Lyapunov's condition** (often with $\delta = 1$) implies (L):

$$\exists \delta > 0 : \quad \lim_n \sum_{k=1}^{n} \frac{\mathbb{E}|X_k - \mathbb{E}[X_k]|^{2+\delta}}{s_n^{2+\delta}} = 0. \tag{Lya}$$

Indeed $|X_k - \mathbb{E}X_k| \ge \varepsilon s_n$ implies $|X_k - \mathbb{E}X_k|^2 \le |X_k - \mathbb{E}X_k|^{2+\delta} / (\varepsilon s_n)^{\delta}$, so $L_n(\varepsilon) \le \varepsilon^{-\delta} \sum_k \mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta} / s_n^{2+\delta}$.

(iii) Uniformly bounded X_n with $s_n \to \infty$ satisfy (Lya) for every $\delta > 0$: if $|X_k| \le \alpha/2$, then $|X_k - \mathbb{E}X_k| \le \alpha$ and

$$\sum_k \frac{\mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta}}{s_n^{2+\delta}} \le \sum_k \frac{\alpha^{\delta} \mathbb{E}|X_k - \mathbb{E}X_k|^2}{s_n^2 s_n^{\delta}} = \Big(\frac{\alpha}{s_n} \Big)^{\delta} \to 0.$$

Lemma 3.57

Lindeberg's condition implies

$$\lim_n \max_{1 \le k \le n} \frac{\sigma_k}{s_n} = 0. \tag{3.14}$$

Proof. For $k \le n$ and $\varepsilon > 0$, splitting the integral on $\{|X_k - \mathbb{E}X_k| \ge \varepsilon s_n\}$:

$$\Big(\frac{\sigma_k}{s_n} \Big)^2 = \int \Big(\frac{X_k - \mathbb{E}X_k}{s_n} \Big)^2 dP \le L_n(\varepsilon) + \varepsilon^2.$$

■

Lemma 3.58

For $t \in \mathbb{R}$, $n \in \mathbb{N}$: $\left|e^{it} - \sum_{k=0}^{n-1} \frac{(it)^k}{k!}\right| \le \frac{|t|^n}{n!}$.

Proof. Taylor's formula with integral remainder for e^{it} (real and imaginary parts) gives the remainder $R_n(t) = \frac{1}{(n-1)!}\int_0^t (t-s)^{n-1} i^n e^{is} ds$, and $|R_n(t)| \le \frac{1}{(n-1)!}\int_0^{|t|} s^{n-1} ds = \frac{|t|^n}{n!}$. ■

Theorem 3.59

For $X \in \mathcal{L}^2$, φ_X is twice continuously differentiable with $\varphi_X'(u) = i\int X e^{iuX} dP$, $\varphi_X''(u) = -\int X^2 e^{iuX} dP$; in particular $\varphi_X'(0) = i\mathbb{E}[X]$, $\varphi_X''(0) = -\mathbb{E}[X^2]$, $|\varphi_X''| \le \mathbb{E}[X^2]$. Moreover

$$\varphi_X(u) = 1 + iu\mathbb{E}[X] + \tfrac{1}{2}\theta(u)u^2\mathbb{E}[X^2] \qquad \text{with } \theta(u) \in \mathbb{C},\ |\theta(u)| \le 1.$$

Proof.

Step 1 (derivatives). Difference quotients of e^{iuX} and iXe^{iuX} are dominated by $|X|$, X^2 (Lemma 3.58, $n = 1$); the DCT gives differentiation under the integral and continuity.

Step 2 (expansion). Lemma 3.58, $n = 2$: $|\varphi_X(u) - 1 - iu\mathbb{E}[X]| \le \frac{1}{2}u^2\mathbb{E}[X^2]$; let $\theta(u)$ be the ratio (or 0). ■

From now on $X_1, X_2, \cdots \in \mathcal{L}^2$ are independent with $\mathbb{E}[X_n] = 0$, $\sigma_n^2 = \mathrm{Var}(X_n) > 0$, $s_n = (\sum_{k\le n}\sigma_k^2)^{1/2}$.

Theorem 3.60

Assume

$$\text{(F)}\quad \lim_n \max_{k\le n} \frac{\sigma_k}{s_n} = 0 \ \text{(Feller)}, \qquad \text{(b)}\quad \lim_n \sum_{k=1}^n \left(\varphi_{X_k}\Big(\frac{u}{s_n}\Big) - 1\right) = -\frac{u^2}{2} \quad \forall u \in \mathbb{R}.$$

Then (X_n) has the CLP.

Proof.

Step 1 (reduction). By Theorem 3.51, $\varphi_{S_n^*}(u) = \prod_{k\le n}\varphi_{X_k}(u/s_n)$. If

$$\lim_n \prod_{k=1}^n \varphi_{X_k}\Big(\frac{u}{s_n}\Big) = e^{-u^2/2} \qquad \forall u, \tag{3.15}$$

then $\varphi_{S_n^*} \to e^{-u^2/2} = \widehat{N(0,1)}$ (Theorem 3.52), which is continuous at 0. Lévy's theorem 3.49 and Theorem 3.47 give $P_{S_n^*} \to N(0,1)$ weakly.

Step 2 (comparison with exponentials). By (b), $\prod_k \exp(\varphi_{X_k}(u/s_n) - 1) \to e^{-u^2/2}$. All factors lie in the unit disc ($|\exp(\varphi_{X_k} - 1)| = \exp(\mathrm{Re}\,\varphi_{X_k} - 1) \le 1$), and $|\prod_k a_k - \prod_k b_k| \le \sum_k |a_k - b_k|$ there, so with $z_k := \varphi_{X_k}(u/s_n) - 1$

$$\Big|\prod_k \varphi_{X_k}\Big(\frac{u}{s_n}\Big) - \prod_k e^{z_k}\Big| \le D_n := \sum_{k=1}^n |z_k + 1 - e^{z_k}|.$$

Step 3. For $\varepsilon \in (0, \frac{1}{2}]$ and $|z| \le \varepsilon$, $|z + 1 - e^z| \le |z|^2 \le \varepsilon|z|$; $|z_k| \le \frac{1}{2}\frac{u^2}{s_n^2}\sigma_k^2 < \varepsilon$ for $n \ge n_0$ (Theorem 3.59, (F)), so

$$D_n \le \varepsilon \sum_k |z_k| \le \varepsilon \frac{u^2}{2} \sum_k \frac{\sigma_k^2}{s_n^2} = \varepsilon \frac{u^2}{2},$$

so $D_n \to 0$. ∎

Proof of Theorem 3.55.

Step 1. We may assume $\mathbb{E}[X_n] = 0$. By Lemma 3.57, (L)⇒(F), so by Theorem 3.60 it remains to derive (b) from (L).

Step 2 (pointwise bounds). Fix u and let

$$Y_k := \Big| e^{i\frac{u}{s_n}X_k} - 1 - i\frac{u}{s_n}X_k + \frac{1}{2}\frac{u^2}{s_n^2}X_k^2 \Big|.$$

Lemma 3.58 ($n = 3$) gives $Y_k \le \frac{1}{6}|\frac{u}{s_n}X_k|^3$, and the triangle inequality with Lemma 3.58 ($n = 2$) gives $Y_k \le \frac{u^2}{s_n^2}X_k^2$.

Step 3. Since $\mathbb{E}[X_k] = 0$ and $\sum_k \sigma_k^2/s_n^2 = 1$,

$$\Big| \sum_{k=1}^{n} \Big(\varphi_{X_k}\Big(\frac{u}{s_n}\Big) - 1 \Big) + \frac{u^2}{2} \Big| = \Big| \sum_k \mathbb{E}\Big[e^{i\frac{u}{s_n}X_k} - 1 - i\frac{u}{s_n}X_k + \frac{u^2}{2s_n^2}X_k^2 \Big] \Big| \le \sum_k \mathbb{E}[Y_k].$$

Step 4. For $\varepsilon > 0$, use the second bound on $\{|X_k| \ge \varepsilon s_n\}$ and the first on the complement, where $\frac{1}{s_n^3}\int_{\{|X_k|<\varepsilon s_n\}} |X_k|^3 dP \le \frac{\varepsilon}{s_n^2}\sigma_k^2$:

$$\sum_k \mathbb{E}[Y_k] \le u^2 \sum_k \int_{\{|X_k/s_n| \ge \varepsilon\}} \Big(\frac{X_k}{s_n}\Big)^2 dP + \frac{|u|^3}{6}\varepsilon \sum_k \frac{\sigma_k^2}{s_n^2} = u^2 L_n(\varepsilon) + \frac{|u|^3}{6}\varepsilon.$$

Hence $\limsup_n \sum_k \mathbb{E}[Y_k] \le \frac{|u|^3}{6}\varepsilon$ for every $\varepsilon > 0$, which proves (b). ∎

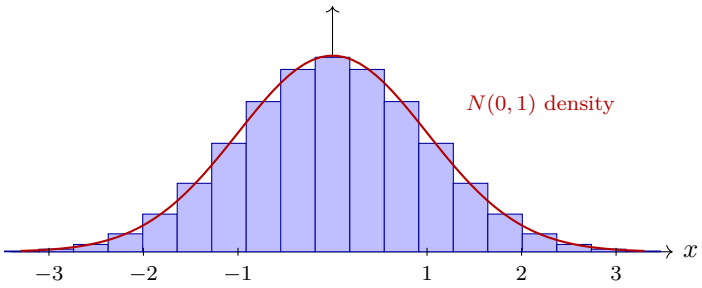


Figure. *Law of S_n^* for $n = 30$ fair tosses (bars of area $P(S_n = k)$) against the $N(0,1)$ density.*

Example 3.61 (Applications)

(i) Ruin probability. n i.i.d. claims $X_i \ge 0$ in $\mathcal{L}^2(m, \sigma^2)$, premium $\Pi := m + \lambda\sigma^2$ ($\lambda > 0$), capital K: ruin $R := \{S_n > K + n\Pi\}$, and the CLT gives

$$P(R) = P\Big(S_n^* > \frac{K + n\lambda\sigma^2}{\sqrt{n}\sigma}\Big) \approx 1 - \Phi(c_n), \qquad c_n := \frac{K + n\lambda\sigma^2}{\sqrt{n}\sigma} \to \infty.$$

c_n increases for $n > K/(\lambda\sigma^2)$: larger portfolios eventually lower the ruin probability.

Remark 3.62

At levels $c_n \to \infty$ the CLT gives no rate; if $\mathbb{E}|X_1 - m|^3 < \infty$, Berry–Esseen (Klenke,

Thm. 15.51) gives

$$\sup_x |P(S_n^* \le x) - \Phi(x)| \le \frac{C}{\sqrt{n}} \frac{\mathbb{E}|X_1 - m|^3}{\sigma^3},$$

which justifies the numbers: with $n = 2000$, $\sigma = 60$, $\lambda = 0.5‰$: (a) $K = 0$ gives $P(R) \approx 1 - \Phi(1.342) \approx 9\%$;
(b) $K = 1500$ gives $P(R) \approx 3\%$. For $P(R) \le 1‰$ with $K = 0$ one needs $\Phi(\sqrt{n}\lambda\sigma) \ge 0.999$, i.e. $n \ge 10\,611$.

(ii) Stirling's formula via the CLT (Weng 1977). Stirling (1730) proved

$$n! \approx \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}, \tag{3.16}$$

used by De Moivre (1733); conversely the CLT (proved without Stirling) yields it. For i.i.d. Poisson(1) X_i, $S_n \sim \pi_n$ with $\mathbb{E}[S_n] = \operatorname{Var}(S_n) = n$ and $S_n^* = t_n(S_n)$, $t_n(x) = \frac{x-n}{\sqrt{n}}$. For $f_\infty(x) := x^- = (-x) \vee 0$, a telescoping sum gives

$$\int f_\infty \, dP_{S_n^*} = e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} \frac{n-k}{\sqrt{n}} = \frac{e^{-n}}{\sqrt{n}} \Big(n + \sum_{k=1}^{n} \Big(\frac{n^{k+1}}{k!} - \frac{n^k}{(k-1)!} \Big) \Big) = \frac{e^{-n} n^{n+\frac{1}{2}}}{n!},$$

while $\int f_\infty \, dN(0,1) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} (-x) e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}}$. So (3.16) follows once

$$\int f_\infty \, dP_{S_n^*} \to \int f_\infty \, dN(0,1). \tag{3.17}$$

f_∞ is unbounded, but $f_m := f_\infty \wedge m \in C_b$ converges by the CLT and $g_m := f_\infty - f_m$ is uniformly small:

$$\int g_m \, dP_{S_n^*} \le \int_{(-\infty, -m]} |x| \, dP_{S_n^*} \le \frac{1}{m} \int x^2 \, dP_{S_n^*} = \frac{1}{m} \operatorname{Var}(S_n^*) = \frac{1}{m},$$

and likewise $\int g_m \, dN(0,1) \le \frac{1}{m}$. A 3ε-argument gives (3.17).